

The physics of vicsek model

Francesco Ginelli. [Eur. Phys. J. Special Topics](#) 225, 2099–2117 (2016)

- Vicsek 模型介绍
- 相图和相变
- 长程有序
- 巨粒子数涨落

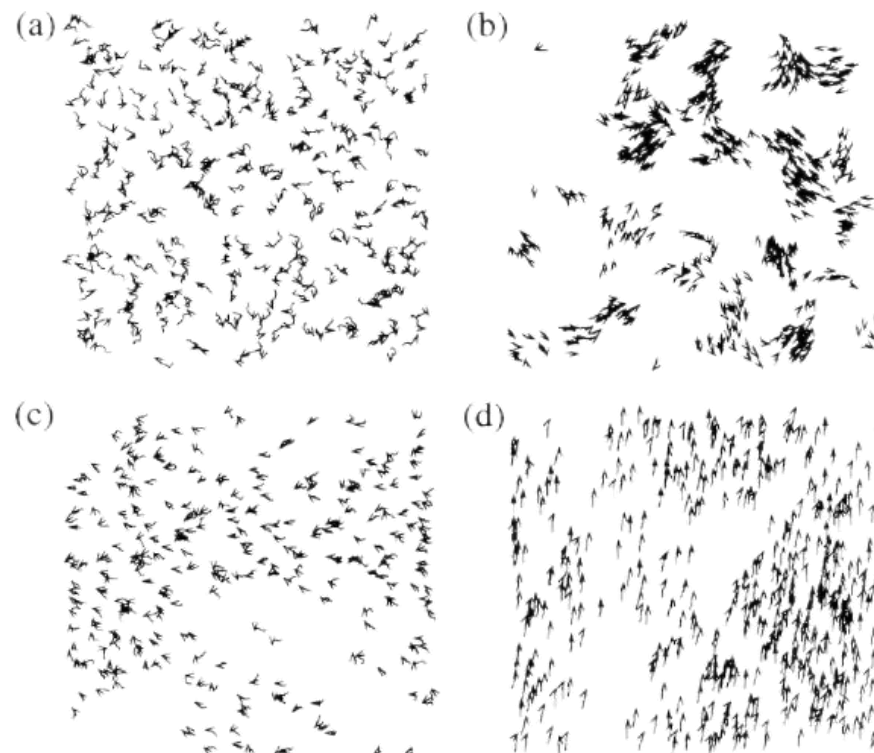
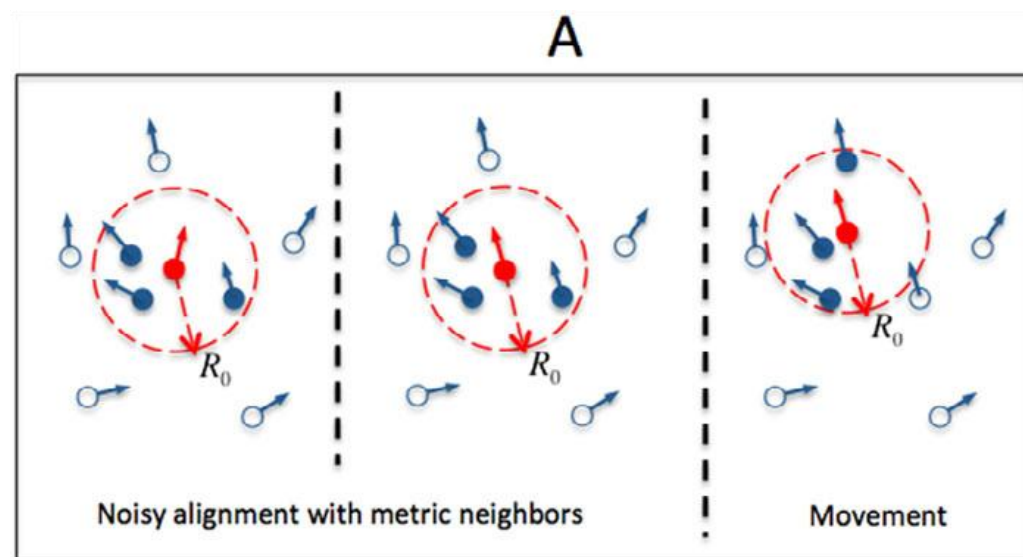
Vicsek模型

$$\vec{v}_i(t + \Delta t) = v_0 (\mathcal{R}_\eta \circ \vartheta) \left[\sum_{j \in \mathcal{S}_i} \vec{v}_j(t) \right]$$

$$\vec{r}_i(t + \Delta t) = \vec{r}_i(t) + \Delta t \vec{v}_i(t + \Delta t) .$$

ϑ 是归一化操作符,

$\mathcal{R}_\eta$ 是围绕目标向量做参数分布的随机旋转



$$\vec{v}_i(t + \Delta t) = v_0 (\mathcal{R}_\eta \circ \vartheta) \left[\sum_{j \in \mathcal{S}_i} \vec{v}_j(t) \right]$$

$$\vec{r}_i(t + \Delta t) = \vec{r}_i(t) + \Delta t \vec{v}_i(t + \Delta t) .$$

参数

总能选取合适的时间和空间度量使得 $\Delta t=1$, $r=1$

维度 d ($=2$) $\mathcal{S}_i$ 半径 r

序参量: $\phi = \langle \varphi(t) \rangle_t$

系统尺寸 L

$$\varphi(t) = |\vec{\varphi}(t)| \quad \vec{\varphi}(t) = \frac{1}{v_0} \langle \vec{v}_i(t) \rangle_i$$

粒子数 $N \rightarrow \frac{n}{L^d} \rightarrow$ 粒子数密度 ρ

控制参量: 噪声强度 η , 粒子数密度 ρ , 运动速率 v_0

$$\phi = \phi(\eta, \rho, v_0)$$

噪声强度 η

运动速率 v_0

步进时间 Δt

相分离和相图

高噪声下:

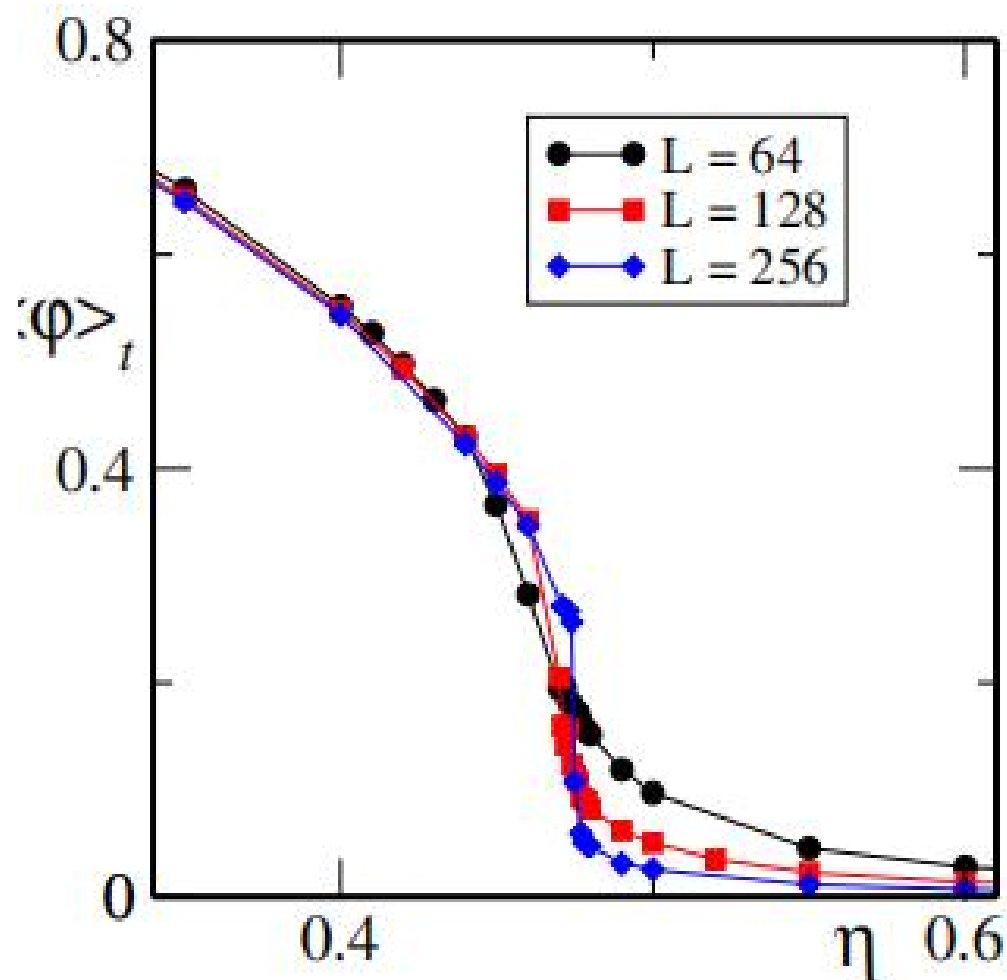
由中心极限定理可得:

$$|\sum_i \vec{v}_i(t)| \sim \sqrt{N} \rightarrow \phi \sim \frac{1}{\sqrt{N}}$$

低噪声下 ($\eta < \eta_c$):

$$\phi \sim 1$$

$$\eta_c = \eta_c(\rho_0, v_0)$$



从信息的传播的角度，对齐相互作用起传递信息的作用，噪声起破坏信息的作用，如果信息能在粒子间传播，那么就能形成长程有序。

对齐相互作用

VS

噪声

把粒子的对齐理解成一种碰撞，有平均自由程

自推进粒子持续长度(SPP persistence length)

$$l \sim \frac{1}{\rho_0}$$

$$l_p \sim \frac{v_0}{\eta^2}$$

开始向有序相转变时: $l \sim l_p$ 可得: $\eta_c \sim \sqrt{\rho_0}$

本末倒置的推导 $l_p \sim \frac{v_0}{\eta^2}$

定义角 θ 为聚合物在位置0的切向量和在距离位置0的距离为 L 处的切向量的夹角。可以证明角的余弦的期望值随距离成指数下降。

$$\langle \cos \theta \rangle = e^{-(L/P)}$$

$$\frac{L}{l_p} = -\ln(\langle \cos \theta \rangle) = -\ln(1 + \langle \cos \theta \rangle - 1) = 1 - \langle \cos \theta \rangle = \frac{1}{2}\theta^2$$

而 $\theta \sim \eta$

有 $l_p \sim \frac{1}{\eta^2}$

相变类型

现在比较公认的是，相变类型受到以下几种因素影响：

- 噪声产生的方式：外部噪声还是内部噪声
- 相互作用方式：度量相互作用还是拓扑相互作用
- $v_0 \Delta t$ 和相互作用半径R的大小关系

噪声产生方式

内部噪声(角度噪声) : 二级相变 (小速度下)

$$\vec{v}_i(t + \Delta t) = v_0 (\mathcal{R}_\eta \circ \vartheta) \left[\sum_{j \in \mathcal{S}_i} \vec{v}_j(t) \right]$$

外部噪声 (矢量噪声) : 一级相变

$$\vec{v}_i(t + \Delta t) = v_0 \vartheta \left[\sum_{j \in \mathcal{S}_i} \vec{v}_j(t) + \eta \mathcal{N}_i \vec{\xi} \right]$$

Grégoire G, Chaté H. Onset of collective and cohesive motion[J]. Physical review letters, 2004, 92(2): 025702.

Aldana M, Dossetti V, Huepe C, et al. Phase transitions in systems of self-propelled agents and related network models[J]. Physical review letters, 2007, 98(9): 095702.

Aldana M, Larralde H, Vázquez B. On the emergence of collective order in swarming systems: a recent debate[J]. International Journal of Modern Physics B, 2009, 23(18): 3661-3685.

相互作用类型

度量相互作用：选取距离自身 R 范围内的粒子作为近邻 二级相变

拓扑相互作用：选取固定 N 个粒子作为近邻，每一步都是这 N 个粒子
一级相变

Ginelli, F. & Chaté, H. Relevance of metric-free interactions in flocking phenomena. Phys. Rev. Lett. 105, 168103 (2010).

$v_0 \Delta t$ 和相互作用半径 R 的大小关系

固定 $\Delta t = 0, = 1$

$v < 0.1$ 二级相变

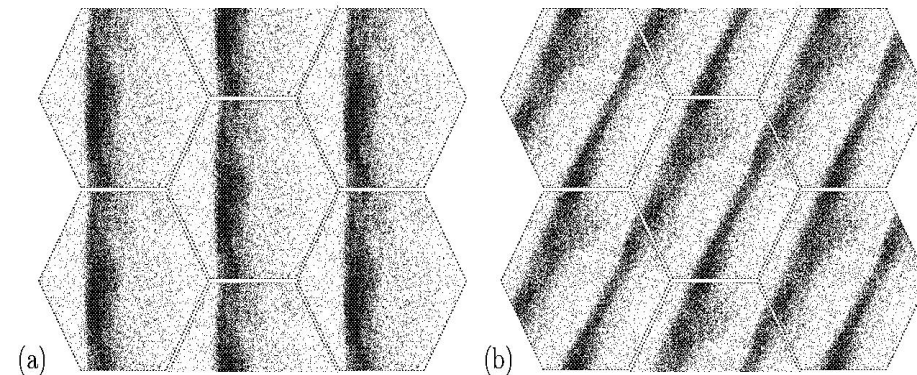
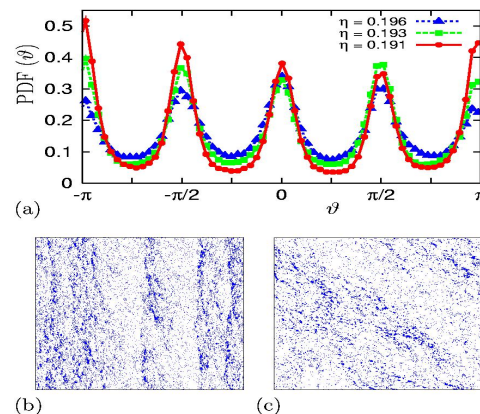
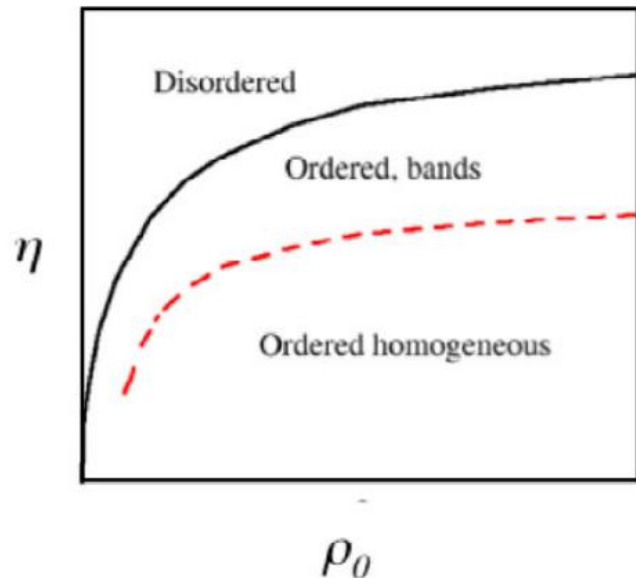
$v > 0.3$ 一级相变

v 非常大 (e.g. $v = 1000$) 二级相变

Nagy M, Daruka I, Vicsek T. New aspects of the continuous phase transition in the scalar noise model (SNM) of collective motion[J]. Physica A: Statistical Mechanics and its Applications, 2007, 373: 445-454.

$$v > 0.3$$

“一级相变”



旅行带的方向取决于边界的对称性

vicsek等人认为旅行带的形成并不是系统的固有属性，而是周期性边界条件的人为因素

在总结中说边界条件的存在，使得我们无法得出关于大速度状态下的物理状态的任何结论

Vicsek-type model	Features	Schematic representation	Observed behavior
original Vicsek Model ⁴³	- point-like particles - intrinsic noise (affecting the final orientation) $\Delta \mathbf{r}(t) \ll R$		
VM with large velocities ⁴⁸	$\Delta \mathbf{r}(t) \gtrsim R$		
metric-free interactions ⁵⁰	communicating with n closest neighbors (a topological rather than a metric representation)		
alternative noise type ^{44, 45}	extrinsic noise (perturbing the individual orientations before averaging)		
bipolar alignment ⁵⁴	choosing the new orientation via original VM or its opposite direction with probability 1/2		- Kosterlitz-Thouless-like transition - giant density fluctuations
VM on a lattice ⁴⁹	- lattice model - volume exclusion		
VM with cohesive forces ^{51,52}	Attractive distance-dependent forces $\mathbf{F}_j$ compete with the VM alignment interaction term to determine the new direction of particle i.		
Examples of purely physical interactions creating collective motion			
volume exclusion ⁶¹ + particle elongation	short-range central attractive forces ⁵⁷	inelastic collisions ⁴⁷	nematic collisions ⁵⁶

Shaebani M R, Wysocki A, Winkler R G, et al. Computational models for active matter[J]. Nature Reviews Physics, 2020, 2(4): 181-199.

长程有序

Mermin-Wagner 定理指出：在二维空间中打破连续对称的系统都不能实现长程序

- xy模型不存在长程序，相变类型是KT相变
- Vicsek模型具有长程序
- Mermin-Wagner 定理只适用于平衡态，而vicsek模型不是平衡态，粒子在移动

Vicsek模型中粒子的移动能力是如何击败Mermin Wagner定理的？

XY模型 (d维)

晶格上所有的自旋都是同一个方向，除了一个“mistaken”自旋，和其他自旋相差 $\delta\theta_0$ ，周围的自旋会去“iron it out”，如果这个“mistake”能够影响到无穷远处的自旋，说明一个小扰动就能破坏长程序，长程序就不存在了

在晶格中，这种传播机制是扩散型的

$$\delta\theta \sim \delta\theta_0/V_e \sim \delta\theta_0\tau^{-\frac{d}{2}}$$

$$\partial_t \delta\theta = \Delta^2 \delta\theta \rightarrow r \sim \sqrt{\tau} \rightarrow V_e \sim \tau^{\frac{d}{2}}$$

噪声波动会不断造成局部误差 (error)，每个自旋的误差和时间成正比

总误差 $n_e \sim \tau V_e$

由中心极限定理 $\Omega_e \sim \sqrt{n_e} \sim \sqrt{\tau V_e}$

每个自旋的误差 $\Delta\theta \sim \Omega_e/V_e \sim \sqrt{\tau/V_e} \sim r^{1-d/2} \rightarrow \begin{cases} 0 & d > 2 \\ \infty & d < 2 \\ \ln L \rightarrow \infty & d = 2. \end{cases}$

Vicsek模型

$$\delta x_{\perp} \sim v_0 \tau \sin \Delta\theta \sim \tau \Delta\theta$$

$$\delta x_{\parallel} \sim v_0 \tau (1 - \cos \Delta\theta) \sim \tau \Delta\theta^2$$

$$V_e \sim w_{\perp}^{d-1}$$

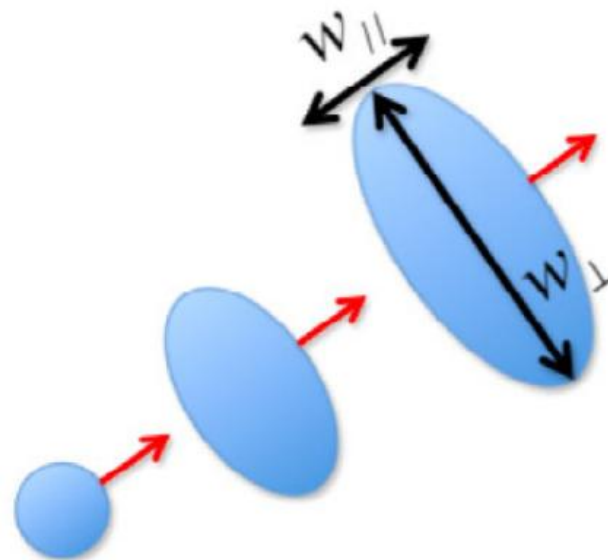
$$\Delta\theta \sim \frac{\tau^{1/2}}{\sqrt{w_{\perp}^{d-1} w_{\parallel}}} \sim \tau^{\gamma}$$



$$2\gamma = 1 - \gamma_{\parallel} - (d-1)\gamma_{\perp}$$

$$\gamma_{\perp} = \max\left(1 + \gamma, \frac{1}{2}\right)$$

$$\gamma_{\parallel} = \max\left(1 + 2\gamma, \frac{1}{2}\right)$$



$$d < 7/3 \quad \gamma = \frac{1-d}{d+3}, \quad \gamma_{\perp} = \frac{4}{d+3} \quad \text{and} \quad \gamma_{\parallel} = \frac{5-d}{d+3} \quad (1)$$

$$7/3 \leq d < 4 \quad \gamma = \frac{3-2d}{2(d+1)}, \quad \gamma_{\perp} = \frac{5}{2(d+1)} \quad \text{and} \quad \gamma_{\parallel} = \frac{1}{2} \quad (2)$$

$$d \geq 4 \quad \gamma = \frac{1}{2} - \frac{d}{4} \quad \text{and} \quad \gamma_{\perp} = \gamma_{\parallel} = \frac{1}{2} \quad (3)$$

Toner&Tu用水动力学重整化群计算表明: $\gamma_{\parallel} = \frac{1}{2}$

因此 (1) 式无效, (2) 式定义域扩充到 $d < 4$

$$d < 4 \quad \gamma = \frac{3-2d}{2(d+1)}, \quad \gamma_{\perp} = \frac{5}{2(d+1)} \quad \text{and} \quad \gamma_{\parallel} = \frac{1}{2}$$

$$d \geq 4 \quad \gamma = \frac{1}{2} - \frac{d}{4} \quad \text{and} \quad \gamma_{\perp} = \gamma_{\parallel} = \frac{1}{2}$$

因此, $d = 2$ 时, $\gamma = -\frac{1}{6} < 0, \Delta\theta \sim \tau^{\gamma}$, 具有长程序

巨粒子数涨落

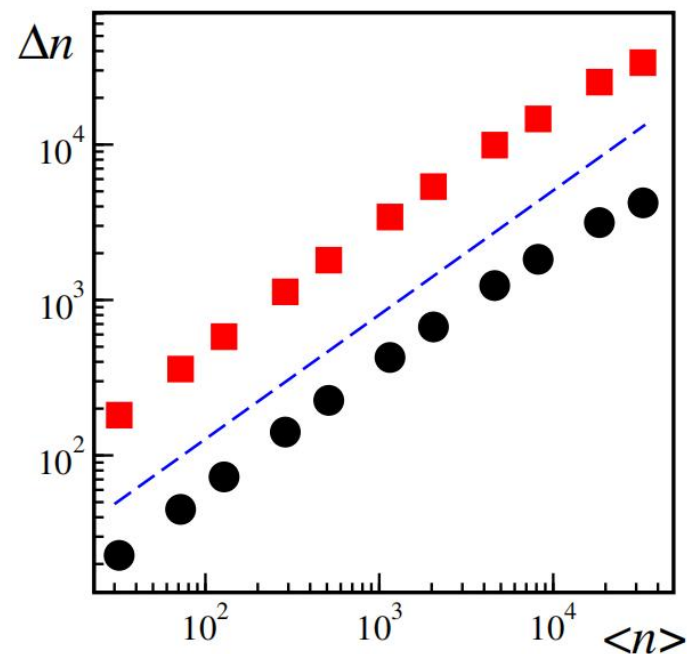
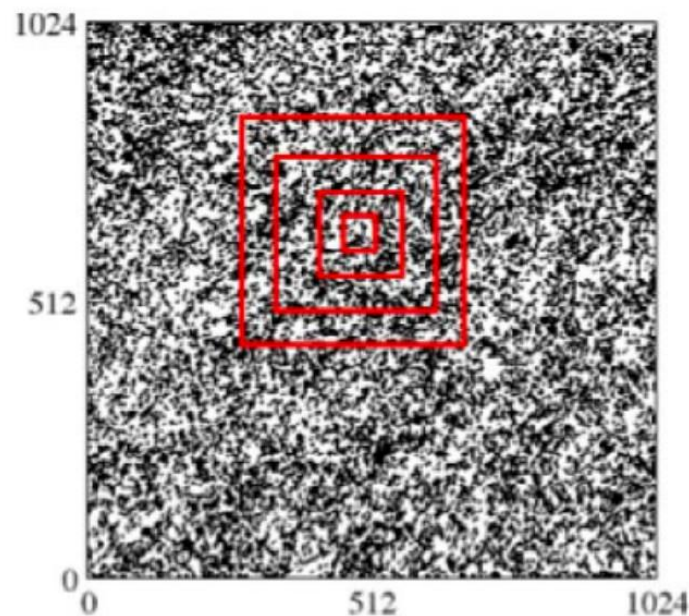
$$\langle n \rangle = \rho_0 \ell^d$$

$$\Delta n = \left(\langle (n_t - \langle n \rangle)^2 \rangle \right)^{1/2}$$

$$\Delta n \sim \langle n \rangle^\alpha$$

中心极限定理里, $\alpha=0.5$

在vicsek模型的有序相中, $\alpha=0.8$
(因为存在长程关联)



结构因子

$$S(\mathbf{q}, t) \equiv \langle \delta \hat{\rho}(\mathbf{q}, t) \delta \hat{\rho}(-\mathbf{q}, t) \rangle \sim \frac{1}{q^\sigma} \quad \text{for } q \rightarrow 0$$

将Toner-Tu方程在系统有序相的稳态附近做线性展开，可以算出结构因子

结构因子和粒子数波动有关系

$$S(\mathbf{q} \rightarrow 0) = \rho_0 \left[\frac{\Delta n^2}{\langle n \rangle} \right]_{n \rightarrow \infty}$$

可以算出

$$\Delta n \sim \langle n \rangle^{1/2 + \sigma/(2d)} \quad \alpha = \frac{1}{2} + \frac{\sigma}{2d} \quad (q \text{ 为有限值时, } \sigma = 0)$$

首先静态结构因子^Q的定义为:

$$S(\mathbf{k}) = \frac{1}{N} \left\langle \sum_{i,j} e^{-i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)} \right\rangle$$
 其实反映的是**密度-密度关联**, 这是因为**单体数密度**^Q

$$n_1(\mathbf{r}) = \langle \sum_i \delta(\mathbf{r} - \mathbf{r}_i) \rangle \text{ 以及两体数密度 } n_2(\mathbf{r}, \mathbf{r}') = \left\langle \sum_{i \neq j} \delta(\mathbf{r} - \mathbf{r}_i) \delta(\mathbf{r}' - \mathbf{r}_j) \right\rangle$$

而根据狄拉克delta函数^Q, 存在关系 $\sum_{i=1}^N e^{i\mathbf{q} \cdot \mathbf{r}_i} = \int d\mathbf{r} e^{-i\mathbf{q} \cdot \mathbf{r}} n(\mathbf{r})$, 于是就可以看到 $S(k)$ 表示其实是**密度关联函数**^Q。在这里不得不说的是, 结构因子和对关联函数的联系同时暗含着, 更深刻的结构因子的意义。

首先如果系统是**平移不变**^Q以及**旋转不变**的, 那么**单体密度**^Q与位置无关, 二体密度只依赖于粒子位置之间的距离, 可以引入对关联函数 $g(r)$, $r = |\mathbf{r} - \mathbf{r}'|$

$$g(r) = n_2(\mathbf{r}, \mathbf{r}') / n^2$$

在热力学和统计物理学^Q中存在一个关系, 即压缩率和涨落的关系,

$$nkT_{\kappa_T} = \frac{\kappa_T}{\kappa_T^{ideal}} = 1 + n \int (g(r) - 1) d\mathbf{r} = \frac{\langle N^2 \rangle - \langle N \rangle^2}{\langle N \rangle}$$

而

$$S(k) = 1 + \frac{N}{V} \int (g(r) - 1) e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r} + \frac{N}{V^2} \left| \int e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r} \right|^2$$

The structure factor [\[edit\]](#)

The second-order correlation function $g^{(2)}(\mathbf{r}_1, \mathbf{r}_2)$ is of special importance, as it is directly related (via a [Fourier transform](#)) to the [structure factor](#) of the system and can thus be determined experimentally using [X-ray diffraction](#) or [neutron diffraction](#).^[3]

If the system consists of spherically symmetric particles, $g^{(2)}(\mathbf{r}_1, \mathbf{r}_2)$ depends only on the relative distance between them, $\mathbf{r}_{12} = \mathbf{r}_2 - \mathbf{r}_1$. We will drop the sub- and superscript: $g(\mathbf{r}) \equiv g^{(2)}(\mathbf{r}_{12})$. Taking particle 0 as fixed at the origin of the coordinates, $\rho g(\mathbf{r}) d^3r = dn(\mathbf{r})$ is the *average* number of particles (among the remaining $N - 1$) to be found in the volume d^3r around the position $\mathbf{r}$.

We can formally count these particles and take the average via the expression $\frac{dn(\mathbf{r})}{d^3r} = \langle \sum_{i \neq 0} \delta(\mathbf{r} - \mathbf{r}_i) \rangle$, with $\langle \cdot \rangle$ the ensemble average, yielding:

$$g(\mathbf{r}) = \frac{1}{\rho} \langle \sum_{i \neq 0} \delta(\mathbf{r} - \mathbf{r}_i) \rangle = V \frac{N-1}{N} \langle \delta(\mathbf{r} - \mathbf{r}_1) \rangle \quad (5)$$

where the second equality requires the equivalence of particles 1, $\dots$, $N - 1$. The formula above is useful for relating $g(\mathbf{r})$ to the static structure factor $S(\mathbf{q})$, defined by $S(\mathbf{q}) = \langle \sum_{ij} e^{-i\mathbf{q}(\mathbf{r}_i - \mathbf{r}_j)} \rangle / N$, since we have:

$$\begin{aligned} S(\mathbf{q}) &= 1 + \frac{1}{N} \langle \sum_{i \neq j} e^{-i\mathbf{q}(\mathbf{r}_i - \mathbf{r}_j)} \rangle = 1 + \frac{1}{N} \left\langle \int_V d\mathbf{r} e^{-i\mathbf{q}\mathbf{r}} \sum_{i \neq j} \delta[\mathbf{r} - (\mathbf{r}_i - \mathbf{r}_j)] \right\rangle \\ &= 1 + \frac{N(N-1)}{N} \int_V d\mathbf{r} e^{-i\mathbf{q}\mathbf{r}} \langle \delta(\mathbf{r} - \mathbf{r}_1) \rangle \end{aligned}$$

, and thus:

$$S(\mathbf{q}) = 1 + \rho \int_V d\mathbf{r} e^{-i\mathbf{q}\mathbf{r}} g(\mathbf{r}), \text{ proving the Fourier relation alluded to above.}$$

This equation is only valid in the sense of [distributions](#), since $g(\mathbf{r})$ is not normalized: $\lim_{r \rightarrow \infty} g(\mathbf{r}) = 1$, so that $\int_V d\mathbf{r} g(\mathbf{r})$ diverges as the volume V , leading to a Dirac peak at the origin for the structure factor. Since this contribution is inaccessible experimentally we can subtract it from the equation above and redefine the structure factor as a regular function:

$$S'(\mathbf{q}) = S(\mathbf{q}) - \rho \delta(\mathbf{q}) = 1 + \rho \int_V d\mathbf{r} e^{-i\mathbf{q}\mathbf{r}} [g(\mathbf{r}) - 1].$$

Finally, we rename $S(\mathbf{q}) \equiv S'(\mathbf{q})$ and, if the system is a liquid, we can invoke its isotropy:

$$S(q) = 1 + \rho \int_V d\mathbf{r} e^{-i\mathbf{q}\mathbf{r}} [g(r) - 1] = 1 + 4\pi\rho \frac{1}{q} \int dr r \sin(qr) [g(r) - 1]. \quad (6)$$

由于对称性破缺，空间各向异性，横向纵向表现出不同的行为，需要对 $S(q)$ 从不同方向求值

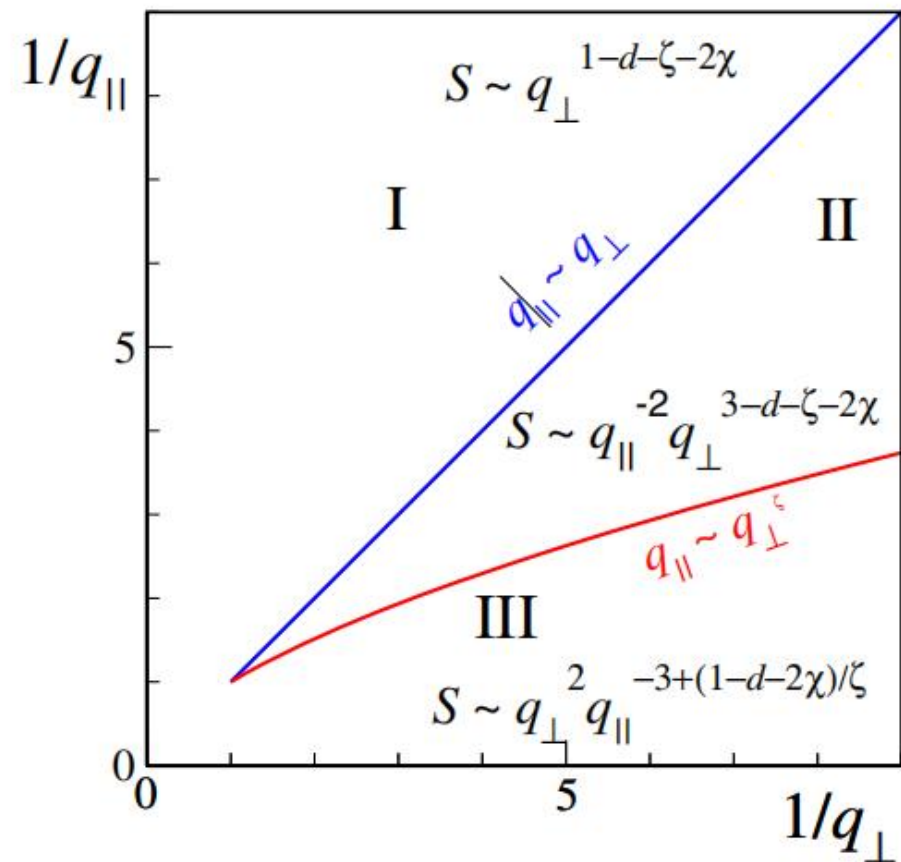
$$S(\mathbf{q}) \sim \begin{cases} q_{\perp}^{1-d-\zeta-2\chi}, & q_{\parallel} \ll q_{\perp} \\ q_{\parallel}^{-2} q_{\perp}^{3-d-\zeta-2\chi}, & q_{\perp}^{\zeta} \gg q_{\parallel} \gg q_{\perp} \\ q_{\parallel}^{-3+(1-d-2\chi)/\zeta} q_{\perp}^2, & q_{\perp}^{\zeta} \ll q_{\parallel} \end{cases}$$

$$3/2 \leq d \leq 4 \quad \chi = \frac{3-2d}{5}, \quad \zeta = \frac{d+1}{5}$$

结构因子主要以 $q_{\parallel} \sim q_{\perp}$ 为主

$$S(\mathbf{q}) \sim q^{-\sigma} \quad \text{with } \sigma = \frac{2}{5}(d+1)$$

$$\alpha = \frac{1}{2} + \frac{d+1}{5d}$$



总结

(1) 相变类型会根据参数的选取方式而改变，比如噪声强度的产生方式、相互作用类型、 vt/R 的大小。

(2) 分别通过信息是否能传到无穷远处和局部的错误是否能传到无穷远处两个角度分析了vicsek模型中的长城有序

(3) 通过“密度关联函数-结构因子-粒子数涨落”的桥梁算出涨落的指数